



HEALTHCONNECT CLINIC — FINAL ANALYTICS & DECISION SUPPORT REPORT

AnalystLab Africa Experience Lab, Week 8

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Week 7-> Week 8 Transition

1. **Main Week 7 output:** A completed Formal Test Log covering 7 tests — validating dashboard calculation accuracy, confirming the combined-risk pattern held robustly across patient segments (Gender, Age Group, Appointment Type), verifying slicer functionality, and statistically testing the Week 6 cross-track feature recommendation.
2. **Most important testing result:** The dashboard's core KPIs and Combined Risk Matrix values were confirmed accurate through independent recalculation, and the combined-risk pattern (64.09% no-show rate for the highest-risk group) proved robust and consistent across every patient segment tested — not an artifact of any single sub-group.
3. **Major issue discovered during testing:** The Week 6 recommendation to Data Science — a single binary high_combined_risk flag — actually underperformed a simpler existing variable (Booking Lead Time alone) when tested statistically, because collapsing four distinct risk groups into two lost meaningful information.
4. **What was refined or improved:** The feature recommendation was revised from a binary flag to a 4-level categorical risk segment feature, which was then retested and confirmed to produce the strongest separation (34.16-point spread) of any feature version tested — a validated, evidence-based improvement over the original.
5. **Which component is now considered ready:** The dashboard's calculation accuracy and slicer functionality are fully validated and ready. The combined-risk finding is validated as robust and ready to present as a core business insight. The revised categorical feature recommendation is statistically validated and ready to hand off.
6. **What still needs to be integrated:** Formal confirmation from the Data Science track that the revised categorical feature was actually incorporated into their model, and whether it improved their model's performance in practice — this remains an open loop since live cross-track contact wasn't achieved.

7. **Track(s) involved in final integration:** Data Science (primary), with findings also relevant to Project Management's final business recommendations coordination.
8. **What I intend to demonstrate during the final presentation:** How the analysis evolved from single-factor KPIs (Week 5) to a validated, tested combined-risk finding (Weeks 6-7) that directly informs both the clinic's intervention strategy and the Data Science track's predictive model — including being transparent about catching and correcting my own earlier recommendation.

Part 1 — Final Integration Readiness

Requirement	Response
What is my final component?	An interactive Power BI dashboard and accompanying analytical report identifying patient no-show risk factors, including a validated combined-risk model showing how factors compound.
What problem does it address?	HealthConnect's core challenge — a 48.46% patient no-show rate — by identifying which specific, measurable factors predict no-shows and which patient segments carry the highest risk.
What is its current status?	Complete and validated. All dashboard calculations have been independently verified against the raw dataset, the combined-risk finding has been tested for consistency across patient segments, and the dashboard's interactivity (slicers) has been confirmed functional.
What did I improve during Week 7?	Corrected a flaw in my own Week 6 feature recommendation — replaced an under-informative binary risk flag with a stronger, statistically validated 4-level categorical risk segment feature.
Which other track(s) does it depend on?	None directly — the Data Analytics track's findings are derived independently from the appointment dataset and Data Dictionary.
Which track(s) depend on my output?	Data Science (for feature engineering in their prediction model) and, indirectly, Project Management (for final business recommendations).
What output will I provide?	A finalized dashboard, validated KPIs, the combined-risk analysis, business insights and recommendations, and a specific, tested feature-engineering specification for the Data Science model.
What output do I need from another track?	Confirmation from Data Science on whether the revised categorical risk feature was incorporated into their model and what effect it had — this remains outstanding due to the lack of live team response.

What evidence demonstrates readiness?	The completed Formal Test Log (7 tests, 6 passes and 1 documented fail-fix-retest cycle), the independently recalculated KPI values matching the dashboard exactly, and the segment-consistency testing confirming the finding's robustness.
What limitations remain?	The dataset is fictional/synthetic, so findings require real-world validation before deployment; no live confirmation was received from Data Science on whether the revised feature recommendation was adopted; the dataset lacks a "reason for no-show" field, limiting root-cause certainty.

Final Analytics & Decision Support Package

1. Final KPIs

The four validated KPIs, confirmed accurate in Week 7 testing:

KPI	RESULT
Overall No-Show Rate	48.46%
No-Show Rate by Reminder Channel (best: SMS)	45.75% (SMS) vs. 51.39% (no reminder)
No-Show Rate by Previous No-Show History	43.51% → 100% (escalating across bands)
No-Show Rate by Booking Lead Time	24.84% → 56.94% (strongest single predictor)
Combined Risk (Validated Week 6-7)	29.94% (lowest) → 64.09% (highest)

2. Validated Findings

- Booking Lead Time is the strongest individual predictor (correlation 0.272).
- Combined risk (prior no-shows + long lead time) more than doubles no-show likelihood (29.94% → 64.09%), and this pattern is robust across gender, age group, and appointment type — confirmed through Week 7 testing.
- SMS reminders outperform other channels, and reminders show a larger benefit specifically within high-risk patients.
- Distance to clinic matters mainly at the extreme (30km+).
- All dashboard calculations independently verified accurate against the raw dataset.

3. Business Insights

1. Nearly half of all appointments are missed — a severe, quantifiable business problem, not a minor inefficiency.

2. Risk factors compound — a patient with both prior history and a long lead time is more than twice as likely to miss their appointment as one with neither.
3. This compounding effect is consistent across every patient demographic tested, meaning it's safe to act on as a general targeting principle.
4. Reminders are most valuable exactly where they're needed most — the high-risk segment.
5. A simple channel switch (prioritizing SMS) is a low-cost, evidence-backed improvement available immediately.

4. Recommendations

1. Prioritize the combined high-risk segment (64.09% no-show rate) for intensive outreach.
2. Shift reminder strategy toward risk-based targeting rather than uniform application.
3. Adopt SMS as the primary reminder channel.
4. Review booking-lead-time policy for long-lead appointments.
5. Investigate distance-based support (e.g., telehealth) for the 30km+ patient segment.

5. Analytical Limitations

The dataset is fictional/synthetic, so findings are illustrative and require real-world validation before deployment. No "reason for no-show" field exists, limiting root-cause certainty. The "5+ previous no-shows" and "30+km distance" segments are based on small samples.

6. Cross-Track Contribution

Provided Data Science with a statistically validated feature specification (a 4-level categorical risk segment, corrected from an initial under-informative binary flag

through Week 7 testing) — a concrete, evidence-based input for their predictive model, arrived at through a genuine test-fix-retest cycle.

7. Executive Summary — HealthConnect Patient No-Show Analytics

HealthConnect Clinic faces a severe patient no-show problem, with nearly half of all appointments (48.46%) going unattended. This analysis, conducted across Weeks 4-7 of the HealthConnect Experience Lab, identified and validated the specific factors driving this problem, moving from initial data exploration through rigorous testing to a final, evidence-backed set of recommendations.

The single strongest predictor of a no-show is how far in advance an appointment is booked: patients booking 22+ days ahead miss their appointment at a rate of 56.94%, more than double the 24.84% rate for near-term bookings. A patient's own history matters too — those with prior no-shows are progressively more likely to miss future appointments. Most significantly, these two factors **compound**: patients carrying both risk factors together show a 64.09% no-show rate, versus just 29.94% for patients with neither — a pattern confirmed, through formal testing, to hold consistently across every patient demographic examined, including gender, age group, and appointment type.

Reminders help, and help most where they're needed most — the high-risk patient segment sees a larger improvement from reminders than lower-risk patients do, and SMS consistently outperforms WhatsApp and Email as a channel.

All findings in this report have been independently verified against the underlying dataset and stress-tested for robustness, including one instance where testing revealed and corrected a flaw in an earlier recommendation — demonstrating not just analytical findings, but a validated, trustworthy analytical process.

Key Recommendation: HealthConnect should prioritize its combined high-risk patients — those with both a prior no-show history and a long booking lead time — for targeted, SMS-first reminder intervention. This single, well-evidenced action addresses the segment responsible for the clinic's most severe attendance risk.

Mandatory HC-POD Final Integration

1. **Track collaborated with:** Data Science
2. **Dependency:** The Data Science track's no-show prediction model depends on validated, well-tested input features rather than assumed-correct ones — this dependency was established in Week 6 and carried through Week 7's testing.
3. **Output received:** The Data Science track's documented target variable (appointment_outcome, specifically the No-Show class) and their general requirement for features that maximize predictive separation, as established in their Week 4-5 documentation.
4. **Output provided:** A final, statistically validated feature specification — a 4-level categorical risk segment (Neither risk factor / Long lead time only / Prior no-shows only / Both factors present) — replacing the initial Week 6 binary flag recommendation after Week 7 testing revealed the binary version lost meaningful information.
5. **Final integration activity:** Completed a full Test → Finding → Action → Retest cycle on my own prior recommendation: tested the binary flag's actual statistical merit, found it underperformed a simpler existing variable, corrected it to a categorical feature, and retested to confirm the correction produced the strongest predictive separation (34.16-point spread) of any feature version evaluated.
6. **What changed:** The Data Science track's candidate feature moved from an under-informative binary flag to a stronger, evidence-validated categorical feature — a materially better technical specification than what was originally delivered.
7. **How the change improved the solution:** A model built on the corrected feature has meaningfully more information to work with in distinguishing risk levels, since the categorical version preserves distinctions the binary

version collapsed away — directly improving the quality of input the Data Science track's model receives.

8. **Evidence:** The Week 7 Formal Test Log entry documenting the comparison across four feature versions (spread and correlation values), and the retest confirmation showing the categorical feature's superior performance — both included in this report's Final Analytics package.
9. **Contribution to the final presentation:** This integration demonstrates the full analytical lifecycle — from initial finding, through cross-track recommendation, to self-tested correction and validated improvement — showing not just what was found, but that the findings were rigorously proven trustworthy before being handed to another track.

Final Presentation

An individual 5–10-minute video presentation covering my track contribution, cross-track collaboration, testing outcomes, and the overall Health Connect solution has been recorded and submitted.

Link:

<https://drive.google.com/drive/folders/1jyOg4gejCZ6g8p34hgOnSCIKoV28zl67?usp=sharing>